嘉定：截止日期是最后 ppt 答辩当天，交给吴助教 2230689@tongji.edu.cn

四平：截止日期是最后 ppt 答辩当天，交给靳助教 2410070@tongji.edu.cn

共两道题。建议交电子版的，但是可也以交纸质的；可以相互讨论，但要自己完成作业。

第一题

根据下面公开可下载的材料，估计中国、美国、欧盟和俄罗斯分别需要种多少棵树，才能抵消（或者中和）它们在 2023 年的碳排放。请先写出你的估算思路，然后再给出关键的计算流程和最终的计算结果。

材料一：《自然》杂志刊登的论文（https://www.nature.com/articles/s43017-024-00532-2.pdf）截图。提示：从截图中找到中国、美国、欧盟和俄罗斯在 2023 年排放了多少吨的碳。注意：1 Gt = 10^9t。



Climate chronicles

https://doi.org/10.1038/s43017-024-00532-2

Global carbon emissions in 2023

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Global CO_2 emissions for 2023 increased by only 0.1% relative to 2022 (following increases of 5.4% and 1.9% in 2021 and 2022, respectively), reaching 35.8 Gt CO_2. These 2023 emissions consumed 10–66.7% of the remaining carbon budget to limit warming to 1.5°C, suggesting permissible emissions could be depleted within 0.5–6 years (67% likelihood).

Annual global CO_2 emissions dropped markedly in 2020 owing to the COVID-19 pandemic, decreasing by 5.8% relative to 2019 (ref. 1). There were hopes that green economic stimulus packages during the COVD crisis might mark the beginning of a longer-term decrease in global emissions toward net-zero emissions, but instead emissions rebounded and quickly exceeded pre-pandemic levels by 2021. However, year-on-year growth has slowed, with 5.4% increases in 2021 (ref. 2) (reaching

At the country level, combined emissions from the top five emitters remain similar to previous years. In descending order, China, the United States, India, the European Union (excluding the UK), and Russia collectively accounted for 64% of global emissions, or 23.0 Gt CO_2. However, interannual fluctuations are apparent when comparing 2022 and 2023, making it difficult to predict long time trends toward zero emissions. For instance, emissions from China (the largest emitter) decreased by 1.9% to 11.0 Gt CO_2 in 2022 but rebounded + 2.9% to 11.3 Gt CO_2 in 2023. By contrast, other regions have maintained earlier increases. Emissions from India, for example, surged by 6.9% to 2.6 Gt CO_2 in 2022 and by another 4.4% to 2.8 Gt CO_2 in 2023; in doing so, India surpassed the EU to become the third highest emitter. Russia exhibited a similar increase, whereby emissions increased by 1.0% to 1.5 Gt CO_2 in 2022 and grew by 2.4% to 1.6 Gt CO_2 in 2023. Meanwhile, emissions began to decrease in other regions. In the United States, emissions increased by 3.0% to 5.0 Gt CO_2 in 2022 but decreased by 2.4% to 4.9 Gt CO_2 in 2023. Similarly, the European Union's emissions increased by 0.3% to 2.8 Gt CO_2 in 2022 but decreased by 6.2% to 2.6 Gt CO_2 in 2023.

材料二：美国农业部官网的科普资料。提示：从材料中找到一棵树木干重的百分之多少是碳？

USDA United States Department of Agriculture

Office of Sustainability & Climate

Forest Carbon FAQs

What is forest carbon?

Carbon in forests comes from carbon dioxide in the atmosphere. This carbon is sometimes called *biogenic* carbon, because it cycles through living organisms. Trees draw carbon dioxide from the atmosphere through a process called photosynthesis. Plants use photosynthesis to produce various carbon-based sugars necessary for tree functioning and to make wood for growth. Every part of a tree stores carbon, from the trunks, branches, leaves, and roots. By weight, dried tree material is about 50 percent carbon. Trees also release carbon dioxide to the atmosphere as a function of their physiology. When some or all parts of a tree decompose after death or burn during fire, the carbon is released back to the atmosphere. Thus, the amount of carbon in forests closely mirrors the natural cycle of tree growth and death.

Carbon can also be found in soils. Carbon in soils comes from the organic matter from trees and other vegetation in varying degrees of decomposition. In fact, soil carbon represents about 50 percent of the total carbon stored in forest systems in the United States. Like vegetation, soils release carbon dioxide when soil microbes break down organic matter. Some soil carbon can decompose in hours or days, but most resides in soils for decades or centuries. In some conditions, carbon resides in soils for thousands of years before fully decomposing. Soil carbon is generally considered very stable, meaning it does not change much or quickly in response to vegetation dynamics. Exceptions are when soils are disturbed significantly, such as tilled for agriculture, with soil erosion, extreme fire events, or with permanent changes in certain types of vegetation cover.

材料三：常见种类的树木的鲜重（green weight）、干重（dry weight）、干重和鲜重的比值和含水量数据。提示：根据下面材料计算常见种类树木的平均鲜重和平均干重和鲜重量比值。注意：1 磅（pound）等于 0.45359237 千克。

Ratio of dry weight to green weight for selected tree species.
The table shows the average weight for one cord of wood. The unit of weight is pounds . Use the ratio of dry weight/green weight to calculate the dry weight of a known standing tree.

Species	green weight	dry weight	ratio d/g	percent water
Apple	4850	3888	0.8016	19.84%
Ash	4184	2880	0.6883	31.17%
Basswood	4404	1984	0.4505	54.95%
Birch	4312	2992	0.6939	30.61%
Boxelder	3589	2632	0.7334	26.66%
Buckeye	4210	1984	0.4713	52.87%

Catalpa	4560	2360	0.5175	48.25%
Cherry, black	3696	2928	0.7922	20.78%
Coffeetree	3872	3112	0.8037	19.63%
Cottonwood	4640	2272	0.4897	51.03%
Douglas-fir	3319	2970	0.8948	10.52%
Elm American	4456	2872	0.6445	35.55%
Elm Red	4800	3112	0.6483	35.17%
Elm Siberian	3800	3020	0.7947	20.53%
Fir, Concolor	3585	2104	0.5869	41.31%
Hackberry	3984	3048	0.7651	23.49%
Hickory,Bitternut	5032	3832	0.7615	23.85%
Hickory, Shagbark	5104	3952	0.7743	22.57%
Honeylocust	4640	3832	0.8259	17.41%
Ironwood	4590	4016	0.8749	12.51%
Juniper,Rocky Mt.	3535	3150	0.8911	10.89%
Locust, Black	4616	4016	0.8700	13.00%
Maple, Other	4685	3680	0.7855	21.45%
Maple, Silver	3940	2752	0.6985	30.15%
Mulbery	4712	3712	0.7878	21.22%
Oak, Bur	4960	3768	0.7597	24.03%
Oak, Red	4888	3528	0.7218	27.82%
Oak, White	5573	4200	0.7536	24.64%
Osage-Orange	5120	4728	0.9234	7.66%
Pine, Eastern White	2780	2250	0.8094	19.06%
Pine, Jack	3200	2488	0.7775	22.25%
Pine,Ponderosa	3600	2336	0.6489	35.11%
Redcedar, Eastern	2959	2632	0.8895	11.05%

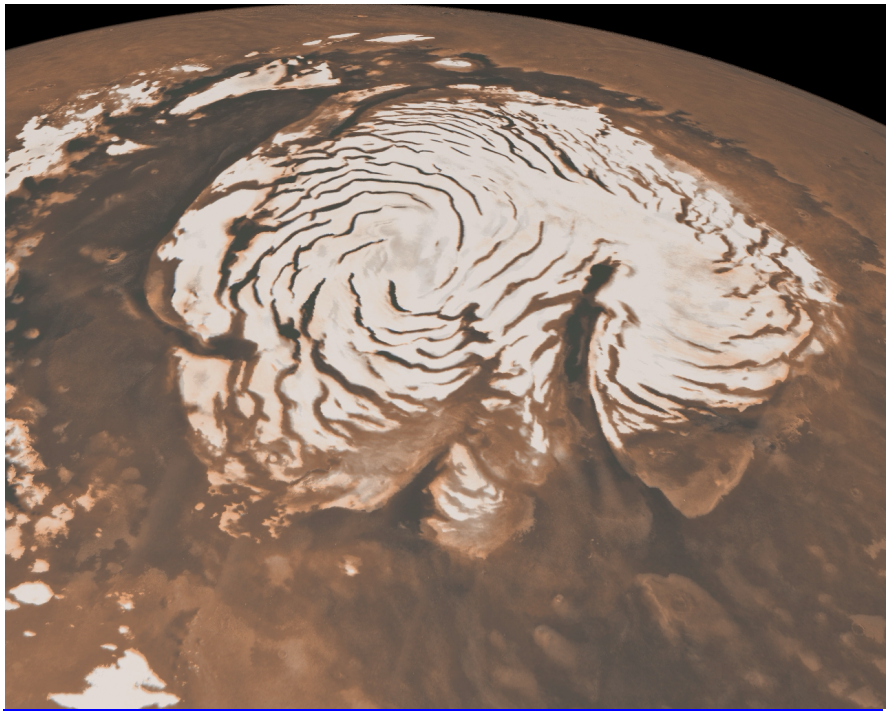
Spruce	2800	2240	0.8000	20.00%
Walnut, Black	4584	3192	0.6963	30.37%
Willow	4320	2540	0.5880	41.20%
Poplar			0.5988	
Sweetgum			0.6526	

The data in this table except Poplar and Sweetgum comes from the Nebraska Extension office.
（http://www.shodor.org/succeedhi/succeedhi/weightree/dryweightcal.html）

第二题

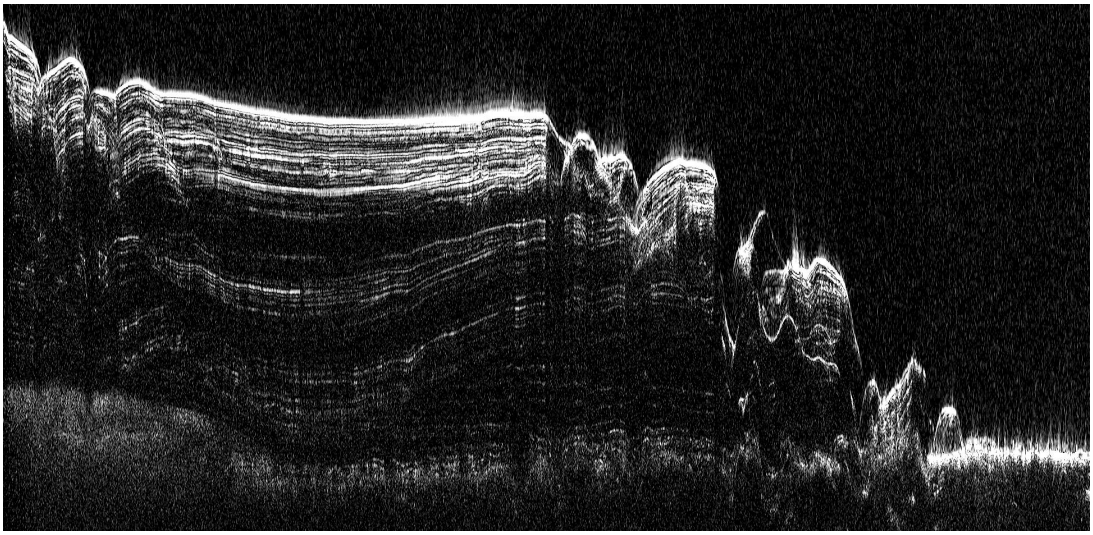
根据图中材料一和材料二中的信息估算火星北极冰盖水量。将您得到的结果与地球海洋中的水量进行比值。提示：地球海水均匀分布在全球表面后的深度为 3.0 km，海水和冰的密度等可以查阅相关资料。

材料 1：火星北极冰盖影像图。提示：从描述资料中找到估计该冰盖半径的数据R。



Northern Ice Cap of Mars. This image, combining data from two instruments aboard NASA's Mars Global Surveyor, depicts an orbital view of the north polar region of Mars.The ice-rich polar cap (the quasi-circular white area at center) is approximately 1,000 kilometers (621 miles) across. The white cap is riven with dark, spiral-shaped bands. These are deep troughs that are in shadow. They do not reflect sunlight as well or have more internal layers exposed. To the right of center, a large canyon, Chasma Boreale, almost bisects the ice cap. (https://science.nasa.gov/resource/northern-ice-cap-of-mars/).

材料 2：火星北极冰盖的剖面图。提示：从描述资料中找到估计该冰盖厚度的数据h。



This image shows a cross-section of a portion of the north polar ice cap of Mars, derived from data acquired by the Mars Reconnaissance Orbiter's Shallow Radar (SHARAD), one of six instruments on the spacecraft. The data depict the region's internal ice structure, with annotations describing different layers. The ice depicted in this graphic is approximately 2 kilometers (1.2 miles) thick and 250 kilometers (155 miles) across. White lines show reflection of the radar signal back to the spacecraft. Each line represents a place where a layer sits on top of another. Scientists study how thick the pancake-like layers are, where they bulge and how they tilt up or down to understand what the surface of the ice sheet was like in the past as each new layer was deposited. (https://science.nasa.gov/resource/north-polar-cap-cross-section/).